中央财经大学 2025–2026 学年第一学期
《偏微分方程》期中试卷参考答案

1. 证. (1) 极坐标与直角坐标满足如下关系

$$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta, \end{cases}$$

于是有

$$\frac{\partial x}{\partial r} = \cos\theta,\ \frac{\partial y}{\partial r} = \sin\theta,\ \frac{\partial x}{\partial \theta} = -r\sin\theta,\ \frac{\partial y}{\partial \theta} = r\cos\theta,$$
$$\frac{\partial^2 x}{\partial r^2} = \frac{\partial^2 y}{\partial r^2} = 0,\ \frac{\partial^2 x}{\partial \theta^2} = -r\cos\theta,\ \frac{\partial^2 y}{\partial \theta^2} = -r\sin\theta,$$

由

$$u_r = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

可得

$$\begin{aligned} u_{rr} &= \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial x}\right)\frac{\partial x}{\partial r} + \frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial r^2} + \frac{\partial}{\partial r}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial r^2} \\ &= \frac{\partial^2 u}{\partial x^2}\left(\frac{\partial x}{\partial r}\right)^2 + 2\frac{\partial^2 u}{\partial x \partial y}\frac{\partial x}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial^2 u}{\partial y^2}\left(\frac{\partial y}{\partial r}\right)^2 + \frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial r^2} + \frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial r^2}. \end{aligned}$$

同理由

$$u_\theta = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

可得

$$u_{\theta\theta} = \frac{\partial^2 u}{\partial x^2}\left(\frac{\partial x}{\partial \theta}\right)^2 + 2\frac{\partial^2 u}{\partial x \partial y}\frac{\partial x}{\partial \theta}\frac{\partial y}{\partial \theta} + \frac{\partial^2 u}{\partial y^2}\left(\frac{\partial y}{\partial \theta}\right)^2 + \frac{\partial u}{\partial x}\frac{\partial^2 x}{\partial \theta^2} + \frac{\partial u}{\partial y}\frac{\partial^2 y}{\partial \theta^2}.$$

故

$$\begin{aligned} &\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} \\ &= \frac{\partial^2 u}{\partial x^2}\left[\left(\frac{\partial x}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial x}{\partial \theta}\right)^2\right] + 2\frac{\partial^2 u}{\partial x \partial y}\left(\frac{\partial x}{\partial r}\frac{\partial y}{\partial r} + \frac{1}{r^2}\frac{\partial x}{\partial \theta}\frac{\partial y}{\partial \theta}\right) \\ &\quad + \frac{\partial^2 u}{\partial y^2}\left[\left(\frac{\partial y}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial y}{\partial \theta}\right)^2\right] + \frac{\partial u}{\partial x}\left(\frac{\partial^2 x}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 x}{\partial \theta^2} + \frac{1}{r}\frac{\partial x}{\partial r}\right) \\ &\quad + \frac{\partial u}{\partial y}\left(\frac{\partial^2 y}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 y}{\partial \theta^2} + \frac{1}{r}\frac{\partial y}{\partial r}\right) \\ &= \frac{\partial^2 u}{\partial x^2}\left(\cos^2\theta + \sin^2\theta\right) + 2\frac{\partial^2 u}{\partial x \partial y}(\cos\theta\sin\theta - \sin\theta\cos\theta) + \frac{\partial^2 u}{\partial y^2}\left(\sin^2\theta + \cos^2\theta\right) \\ &\quad + \frac{\partial u}{\partial x}\left(-\frac{1}{r}\cos\theta + \frac{1}{r}\cos\theta\right) + \frac{\partial u}{\partial y}\left(-\frac{1}{r}\sin\theta + \frac{1}{r}\sin\theta\right) \\ &= u_{xx} + u_{yy}. \end{aligned}$$

(2) 是调和函数.

直接计算可得

$$v_r = nr^{n-1}\cos n\theta,\ v_{rr} = n(n-1)r^{n-2}\cos n\theta,\ v_{\theta\theta} = -n^2r^n\cos n\theta$$

故由 (1) 知

$$\Delta v = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta} = n(n-1)r^{n-2}\cos n\theta + nr^{n-2}\cos n\theta - n^2r^{n-2}\cos n\theta = 0.$$

□

2. 解. 对 $\forall v \in V, \lambda \in \mathbb{R}$, 有

$$\begin{aligned}
&J(u+\lambda v)\\
=&\frac{1}{2}\iint_\Omega \left[|\nabla u + \lambda\nabla v|^2 + c(u+\lambda v)^2\right]\mathrm{d}x\mathrm{d}y - \iint_\Omega f\cdot(u+\lambda v)\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} g\cdot(u+\lambda v)\mathrm{d}s\\
=&\frac{1}{2}\iint_\Omega \left(|\nabla u|^2 + cu^2\right)\mathrm{d}x\mathrm{d}y - \iint_\Omega fu\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} gu\mathrm{d}s\\
&+\lambda\left[\iint_\Omega(\nabla u\cdot\nabla v + cuv - fv)\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} gv\mathrm{d}s\right]\\
&+\frac{1}{2}\lambda^2\iint_\Omega\left(|\nabla v|^2 + cv^2\right)\mathrm{d}x\mathrm{d}y.
\end{aligned}$$

由于 $J(u+\lambda v)$ 在 $\lambda = 0$ 处取到最小值, 故

$$0 = \left.\frac{\mathrm{d}J(u+\lambda v)}{\mathrm{d}\lambda}\right|_{\lambda=0} = \iint_\Omega(\nabla u\cdot\nabla v + cuv - fv)\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} gv\mathrm{d}s.$$

由 Green 公式知对 $\forall v \in V$ 有

$$\iint_\Omega(-\Delta u + cu - f)v\mathrm{d}x\mathrm{d}y + \int_{\partial\Omega}\left(\frac{\partial u}{\partial\vec{n}} - g\right)v\mathrm{d}s = 0.$$

下先证 $-\Delta u + cu = f$ 在 Ω 上恒成立. 反证而设存在 $M_0 \in \Omega$, 使得 $(-\Delta u + cu - f)(M_0) \neq 0$. 不始设 $(-\Delta u + cu - f)(M_0) > 0$. 由 $u \in C^2(\Omega)$ 与 $f \in C(\Omega)$ 知 $-\Delta u + cu - f \in C(\Omega)$, 故存在 M_0 的邻域 $U_1 \subset \Omega$, 使得 $(-\Delta u + cu - f)|_{U_1} > 0$. 取 M_0 的邻域 $U_2 \subsetneqq U_1$, 并取 $v \in C^2(\Omega)\cap C^1(\bar{\Omega})$ 使得 $v|_{U_1} \geqslant 0, v|_{U_2} > 0, v|_{\bar{\Omega}\setminus U_1} = 0$. 于是有

$$\begin{aligned}
&\iint_\Omega(-\Delta u + cu - f)v\mathrm{d}x\mathrm{d}y + \int_{\partial\Omega}\left(\frac{\partial u}{\partial\vec{n}} - g\right)v\mathrm{d}s\\
=&\iint_{U_1}(-\Delta u + cu - f)v\mathrm{d}x\mathrm{d}y\\
\geqslant&\iint_{U_2}(-\Delta u + cu - f)v\mathrm{d}x\mathrm{d}y\\
>&\,0.
\end{aligned}$$

矛盾, 故 $-\Delta u + cu = f$ 在 Ω 上恒成立. 于是有

$$\int_{\partial\Omega}\left(\frac{\partial u}{\partial\vec{n}} - g\right)v\mathrm{d}s = 0, \quad \forall v \in V.$$

同理, 利用 $v \in C^1(\bar{\Omega})$ 的任意性知

$$\left.\left(\frac{\partial u}{\partial \vec{n}} - g\right)\right|_{\partial\Omega} = 0.$$

综上, 有

$$\begin{cases} -\Delta u + cu = f & \text{在 } \Omega \text{ 上}, \\ \left.\dfrac{\partial u}{\partial \vec{n}}\right|_{\partial\Omega} = g. \end{cases} \tag{1}$$

此即为由题设中变分问题导出的边值问题.

下证该边值问题的解即为题设中变分问题的解, 这就证明了等价性.

设 $u \in V$ 是边值问题 (1) 的解. 对 $\forall v \in V$, 记 $w = v - u$. 则有

$$\begin{aligned} J(v) = J(u+w) = J(u) &+ \left[\iint_\Omega (\nabla u \cdot \nabla w + cuw - fw)\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} gw\mathrm{d}s\right] \\ &+ \frac{1}{2}\iint_\Omega \left(|\nabla w|^2 + cw^2\right)\mathrm{d}x\mathrm{d}y. \end{aligned}$$

由 (1) 以及 Green 公式知

$$\begin{aligned} &\iint_\Omega (\nabla u \cdot \nabla w + cuw - fw)\mathrm{d}x\mathrm{d}y - \int_{\partial\Omega} gv\mathrm{d}s \\ =& \iint_\Omega (-\Delta u + cu - f)w\mathrm{d}x\mathrm{d}y + \int_{\partial\Omega}\left(\frac{\partial u}{\partial \vec{n}} - g\right)w\mathrm{d}s \\ =& \, 0. \end{aligned}$$

故 $J(v) \geqslant J(u)$ 对 $\forall v \in V$ 成立, 即 $J(u) = \min\limits_{v \in V} J(v)$. 证毕. □

3. 解. 圆盘 $B(O,R)$ 上泊松公式为

$$\begin{aligned} u(\rho_0, \theta_0) &= \frac{R^2 - \rho_0^2}{2\pi}\int_0^{2\pi} \frac{f(\theta)}{R^2 - 2R\rho_0\cos(\theta - \theta_0) + \rho_0^2}\mathrm{d}\theta \\ &= \frac{R^2 - \rho_0^2}{2\pi}\int_0^{2\pi} \frac{f(\theta + \theta_0)}{R^2 - 2R\rho_0\cos\theta + \rho_0^2}\mathrm{d}\theta, \end{aligned}$$

其中 $f(\theta) = 2 + \sin 3\theta$.

取 $z = \dfrac{\rho_0}{R}e^{\mathrm{i}\theta}\left(\dfrac{\rho_0}{R} < 1\right)$, 则可得

$$\frac{R^2 - \rho_0^2}{R^2 - 2R\rho_0\cos\theta + \rho_0^2} = \frac{1 - \left(\frac{\rho_0}{R}\right)^2}{1 - 2\frac{\rho_0}{R}\cos\theta + \left(\frac{\rho_0}{R}\right)^2} = \frac{1 - z\bar{z}}{1 - z - \bar{z} + z\bar{z}} = 1 + 2\sum_{n=1}^{+\infty}\left(\frac{\rho_0}{R}\right)^n \cos n\theta.$$

另一方面, 注意到 $\{1\}\cup\{\sin n\theta,\cos n\theta\}_{n=1}^{\infty}$ 关于 $L^2(0,2\pi)$ 内积构成正交系. 因此有

$$\begin{aligned}
u(\rho_0,\theta_0) &= \frac{1}{2\pi}\int_0^{2\pi}\frac{(R^2-\rho_0^2)[2+\sin(3\theta+3\theta_0)]}{R^2-2R\rho_0\cos\theta+\rho_0^2}\mathrm{d}\theta \\
&= 2+\frac{1}{2\pi}\int_0^{2\pi}(\sin 3\theta\cos 3\theta_0+\cos 3\theta\sin 3\theta_0)\left[1+2\sum_{n=1}^{+\infty}\left(\frac{\rho_0}{R}\right)^n\cos n\theta\right]\mathrm{d}\theta \\
&= 2+\frac{1}{\pi}\left(\frac{\rho_0}{R}\right)^3\sin 3\theta_0\int_0^{2\pi}\cos^2 3\theta\mathrm{d}\theta \\
&= 2+\frac{1}{\pi}\left(\frac{\rho_0}{R}\right)^3\sin 3\theta_0\int_0^{2\pi}\frac{1+\cos 6\theta}{2}\mathrm{d}\theta \\
&= 2+\left(\frac{\rho_0}{R}\right)^3\sin 3\theta_0.
\end{aligned}$$

□

4. 解. 将原初边值问题拆为如下两个初边值问题:

$$\begin{cases}
u_t=a^2u_{xx} \quad (t>0, 0<x<2), \\
u(x,0)=\begin{cases} x, & 0<x\leqslant 1, \\ 2-x, & 1<x<2, \end{cases} \\
u(0,t)=u(2,t)=0 \quad (t>0)
\end{cases} \tag{2}$$

和

$$\begin{cases}
u_t=a^2u_{xx}+\sin(\pi x)e^t \quad (t>0, 0<x<2), \\
u(x,0)=0 \quad (0<x<2), \\
u(0,t)=u(2,t)=0 \quad (t>0).
\end{cases} \tag{3}$$

若 u_1 是 (2) 的解, u_2 是 (3) 的解, 则 $u=u_1+u_2$ 是原初边值问题的解.

先来求 u_1. 令 $u_1(x,t)=X(x)T(t)$, 则有

$$\begin{cases}
T'(t)+\lambda a^2T(t)=0, \\
X''(x)+\lambda X(x)=0.
\end{cases}$$

由边界条件知 $X(0)T(t)=X(2)T(t)=0$, 即 $X(0)=X(2)=0$. 由关于 X 的方程得

$$X(x)=\begin{cases}
Ae^{\sqrt{-\lambda}x}+Be^{-\sqrt{-x}x}, & \lambda<0, \\
A+Bx, & \lambda=0, \\
A\cos\sqrt{\lambda}x+B\sin\sqrt{\lambda}x, & \lambda>0.
\end{cases}$$

由边界条件知 $\lambda\leqslant 0$ 时只有零解. 当 $\lambda>0$ 时, 我们得

$$\begin{cases}
A=0, \\
B\sin(2\sqrt{\lambda})=0.
\end{cases}$$

为使 $B \neq 0$, 需 $\sin(2\sqrt{\lambda}) = 0$. 故 $2\sqrt{\lambda} = k\pi$, $k = 1, 2, \ldots$ 因此 $\lambda_k = \left(\frac{k\pi}{2}\right)^2$, $k = 1, 2, \ldots$ 相应地,

$$X_k = B_k \sin\left(\frac{k\pi}{2}x\right),\ k = 1, 2, \ldots$$

由 T 的方程知

$$T_k(t) = C_k e^{-\frac{a^2k^2\pi^2}{4}t},\ k = 1, 2, \ldots$$

于是得到题中齐次方程齐次 Dirichlet 边界条件的级数解为

$$u_1(x,t) = \sum_{k=1}^{+\infty} A_k e^{-\frac{a^2k^2\pi^2}{4}t} \sin\left(\frac{k\pi}{2}x\right),$$

其中

$$\begin{aligned}
A_k &= \int_0^2 u(x,0)\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x \\
&= \int_0^1 x\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x + \int_1^2 (2-x)\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x \\
&= \int_0^1 x\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x + \int_1^0 x\sin\left(k\pi - \frac{k\pi}{2}x\right)\mathrm{d}(2-x) \\
&= \int_0^1 x\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x + (-1)^{k+1}\int_0^1 x\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x \\
&= \begin{cases} 2\displaystyle\int_0^1 x\sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x, & k \text{ 为奇数}, \\ 0, & k \text{ 为偶数}. \end{cases}
\end{aligned}$$

直接计算可得

$$\begin{aligned}
\int_0^1 x\sin\left(\frac{2k+1}{2}\pi x\right)\mathrm{d}x &= -\frac{2}{(2k+1)\pi}\int_0^1 x\mathrm{d}\cos\left(\frac{2k+1}{2}\pi x\right) \\
&= -\frac{2}{(2k+1)\pi}\left[x\cos\left(\frac{2k+1}{2}\pi x\right)\Big|_0^1 - \int_0^1 \cos\left(\frac{2k+1}{2}\pi x\right)\mathrm{d}x\right] \\
&= \frac{4}{(2k+1)^2\pi^2}\sin\left(\frac{2k+1}{2}\pi x\right)\Big|_0^1 \\
&= (-1)^k\frac{4}{(2k+1)^2\pi^2}.
\end{aligned}$$

因此 $A_{2k} = 0$, $A_{2k+1} = (-1)^k\frac{8}{(2k+1)^2\pi^2}, k = 0, 1, 2, \ldots$ 故

$$u_1(x,t) = \sum_{k=0}^{+\infty}(-1)^k\frac{8}{(2k+1)^2\pi^2}e^{-\frac{a^2(2k+1)^2\pi^2}{4}t}\sin\left(\frac{2k+1}{2}\pi x\right). \tag{4}$$

下面求 u_2. 则由齐次化原理可知

$$u_2(x,t) = \int_0^t w(x,t;\tau)\mathrm{d}\tau,$$

其中 $w(x,t;\tau)$ 满足

$$\begin{cases} w_t = a^2 w_{xx} \quad (t>\tau, 0<x<2), \\ w(x,\tau;\tau) = \sin(\pi x)e^{\tau} \quad (0<x<2), \\ w(0,t;\tau) = w(2,t;\tau) = 0 \quad (t>\tau). \end{cases}$$

令 $t' = t-\tau$. 则

$$\begin{cases} w_{t'} = a^2 w_{xx} \quad (t'>0, 0<x<2), \\ t'=0: w = \sin(\pi x)e^{\tau} \quad (0<x<2), \\ x=0 \text{ 及 } x=2: w=0 \quad (t'>0). \end{cases}$$

于是由上述求解过程知

$$w(x,t;\tau) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2k^2\pi^2}{4}t'} \sin\left(\frac{k\pi}{2}x\right) = \sum_{k=1}^{+\infty} D_k(\tau) e^{-\frac{a^2k^2\pi^2}{4}(t-\tau)} \sin\left(\frac{k\pi}{2}x\right),$$

其中

$$D_k(\tau) = \int_0^2 \sin(\pi x)e^{\tau} \sin\left(\frac{k\pi}{2}x\right)\mathrm{d}x = \begin{cases} e^{\tau}, & k=2, \\ 0, & \text{其它}. \end{cases}$$

故

$$w(x,t;\tau) = e^{-a^2\pi^2 t}\sin(\pi x)e^{\left(1+a^2\pi^2\right)\tau}.$$

因此

$$\begin{aligned} u_2(x,t) &= \int_0^t w(x,t;\tau)\mathrm{d}\tau \\ &= e^{-a^2\pi^2 t}\sin(\pi x)\int_0^t e^{\left(1+a^2\pi^2\right)\tau}\mathrm{d}\tau \\ &= \frac{\sin(\pi x)}{1+a^2\pi^2}(e^t - e^{-a^2\pi^2 t}). \end{aligned} \tag{5}$$

于是由 (4) 与 (5) 知原初边值问题的解为

$$\begin{aligned} u(x,t) &= u_1(x,t) + u_2(x,t) \\ &= \frac{\sin(\pi x)}{1+a^2\pi^2}(e^t - e^{-a^2\pi^2 t}) + \sum_{k=0}^{+\infty}(-1)^k \frac{8}{(2k+1)^2\pi^2} e^{-\frac{a^2(2k+1)^2\pi^2}{4}t} \sin\left(\frac{2k+1}{2}\pi x\right). \end{aligned}$$

□

5. 解. 记

$$f_1(x) = \begin{cases} 2, & |x| \le 3, \\ 0, & x>3 \end{cases}$$

则 $f(x)=x^4 f_1(x)$. 于是有

$$
\begin{aligned}
\hat{f}_1(\lambda) &= \int_{-\infty}^{+\infty} f_1(x)e^{-\mathrm{i}\lambda x}\mathrm{d}x \\
&= \int_{-3}^{3} 2(\cos\lambda x - \mathrm{i}\sin\lambda x)\mathrm{d}x \\
&= 4\int_{0}^{3} \cos\lambda x\mathrm{d}x \\
&= \frac{4\sin 3\lambda}{\lambda} \\
&= 4\sum_{n=0}^{+\infty}(-1)^n\frac{3^{2n+1}\lambda^{2n}}{(2n+1)!}
\end{aligned}
$$

从而有

$$
\begin{aligned}
\hat{f}(\lambda) &= F[x^4 f_1](\lambda) \\
&= \frac{1}{(-\mathrm{i})^4}\hat{f}_1^{(4)}(\lambda) \\
&= 4\sum_{n=2}^{+\infty}(-1)^n\frac{3^{2n+1}}{2n+1}\frac{\lambda^{2n-4}}{(2n-4)!} \\
&= 4\sum_{n=0}^{+\infty}(-1)^n\frac{3^{2n+5}}{2n+5}\frac{\lambda^{2n}}{(2n)!} \\
&= 972\sum_{n=0}^{+\infty}(-1)^n\frac{(3\lambda)^{2n}}{(2n+5)(2n)!}.
\end{aligned}
$$

或者直接计算 $\hat{f}_1^{(4)}(\lambda)$ 有

$$
\hat{f}(\lambda)=\hat{f}_1^{(4)}(\lambda)=\left(\frac{96}{\lambda^5}-\frac{432}{\lambda^3}+\frac{324}{\lambda}\right)\sin 3\lambda+\left(\frac{432}{\lambda^2}-\frac{288}{\lambda 4}\right)\cos 3\lambda.
$$

□

6. 解. 极值原理: 对不恒等于常数的调和函数 $u(x,y,z)$, 其在开区域 Ω 的任何内点上的值不可能达到它在 Ω 上的上界或下界.

通过该极值原理, 可以证明狄利克雷边界条件下 Poisson 方程解的唯一性以及解对边界条件的连续依赖性. □